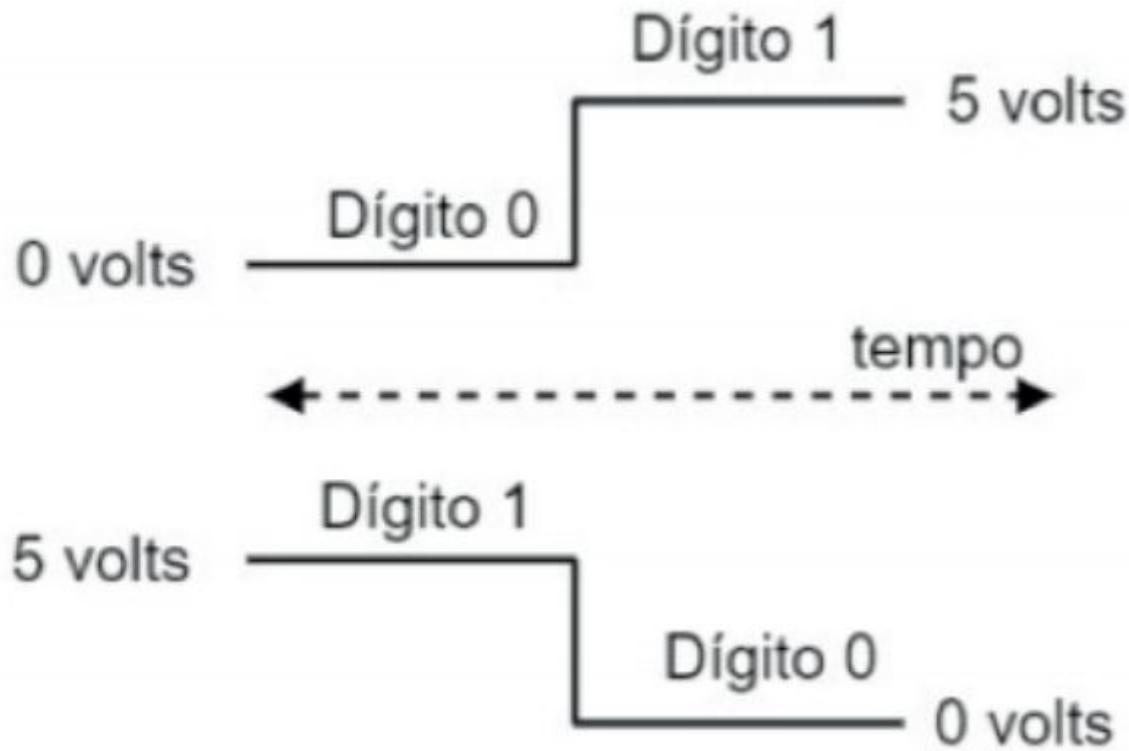
Circuitos Digitais

Nível lógico

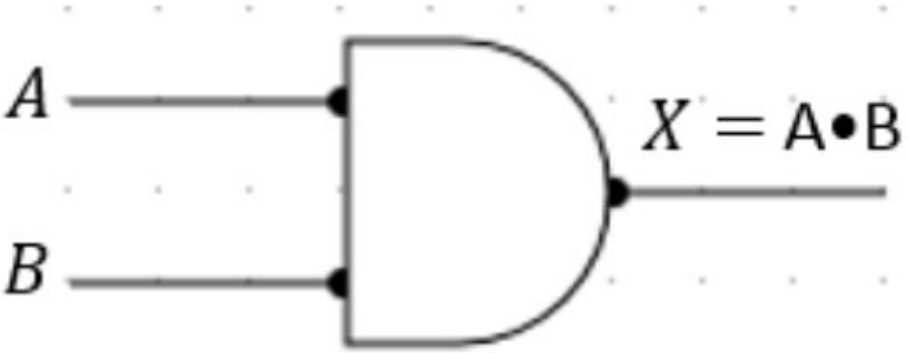


“E”

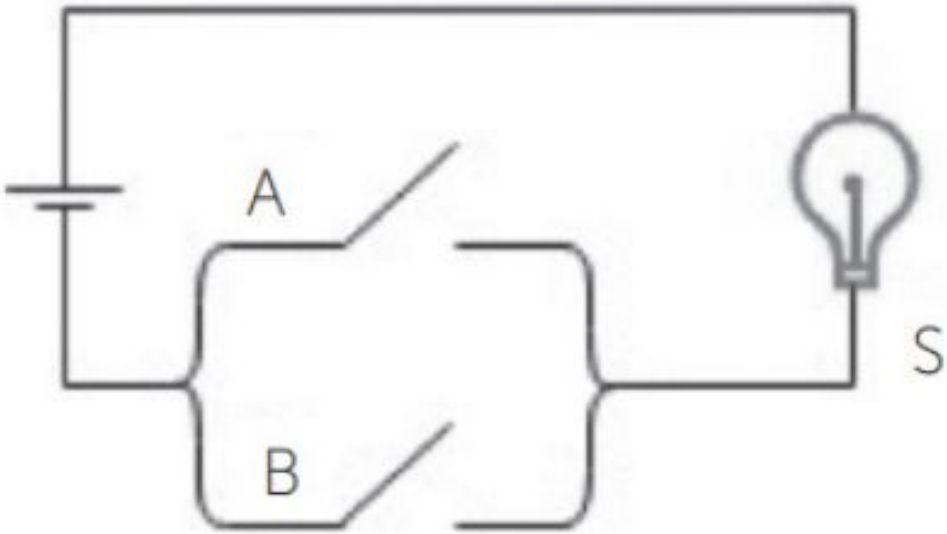
PORTA AND

A	B	$X = A \bullet B$
0	0	0
0	1	0
1	0	0
1	1	1

SÍMBOLO DE UMA PORTA AND



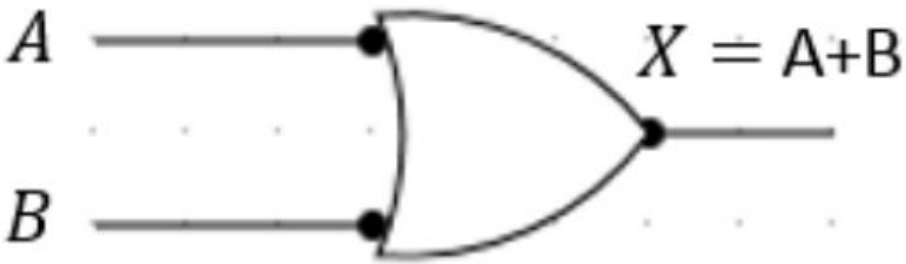
"OU"



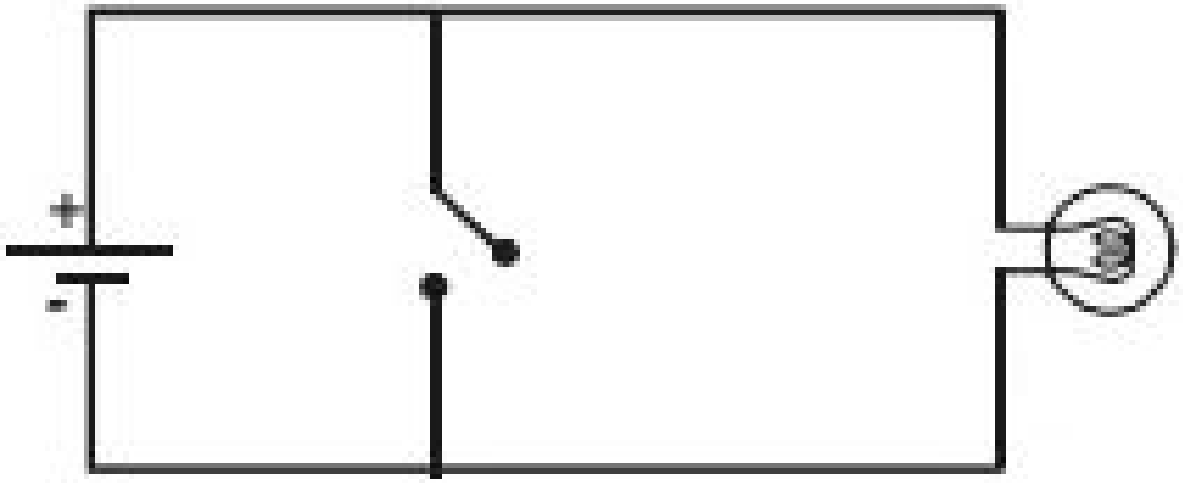
PORTA OR

A	B	$X = A+B$
0	0	0
0	1	1
1	0	1
1	1	1

SÍMBOLO DE UMA PORTA OR



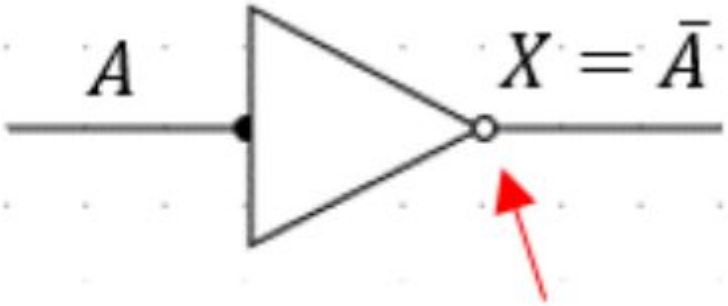
“NÃO”



INVERSOR

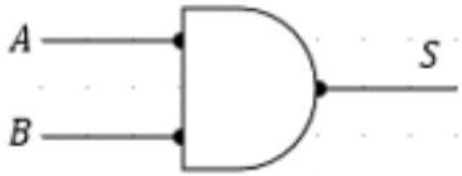
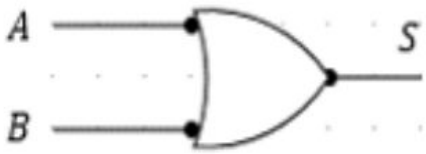
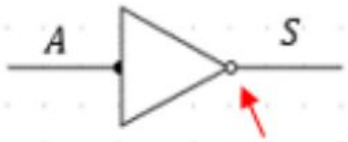
A	$X = \bar{A}$
0	1
1	0

SÍMBOLO DE UM INVERSOR

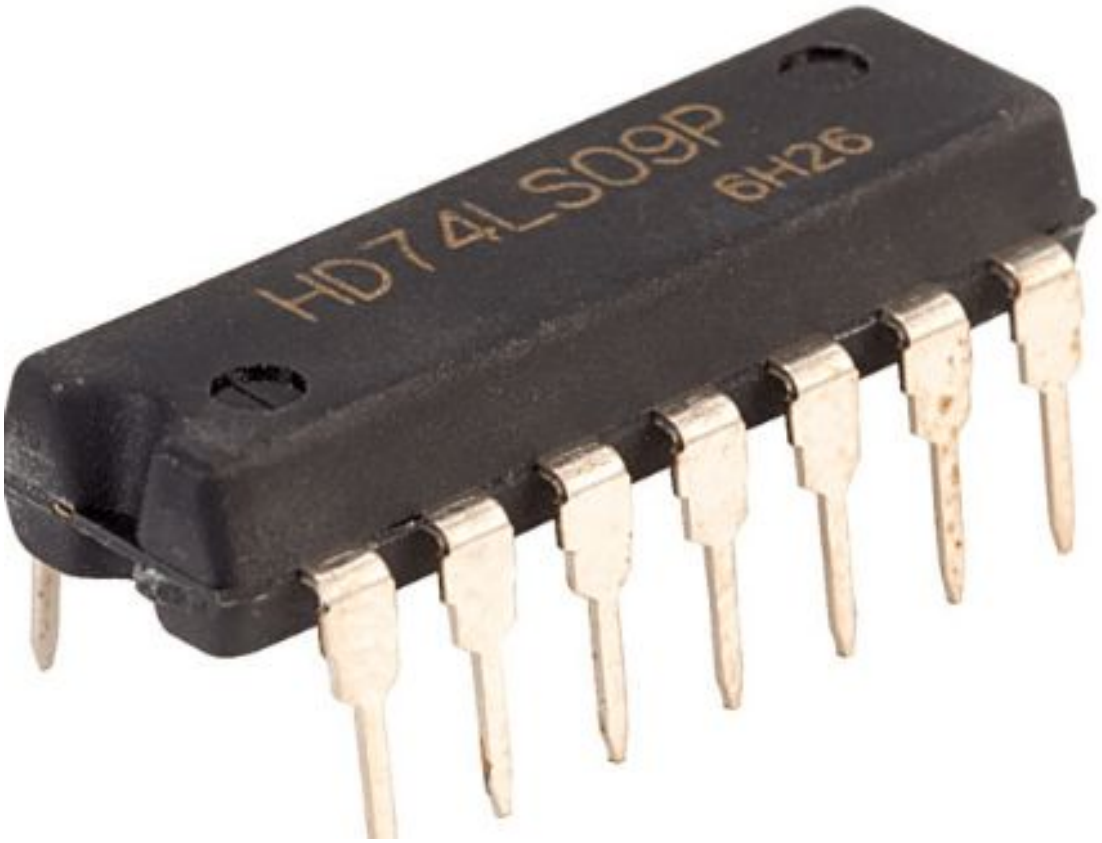


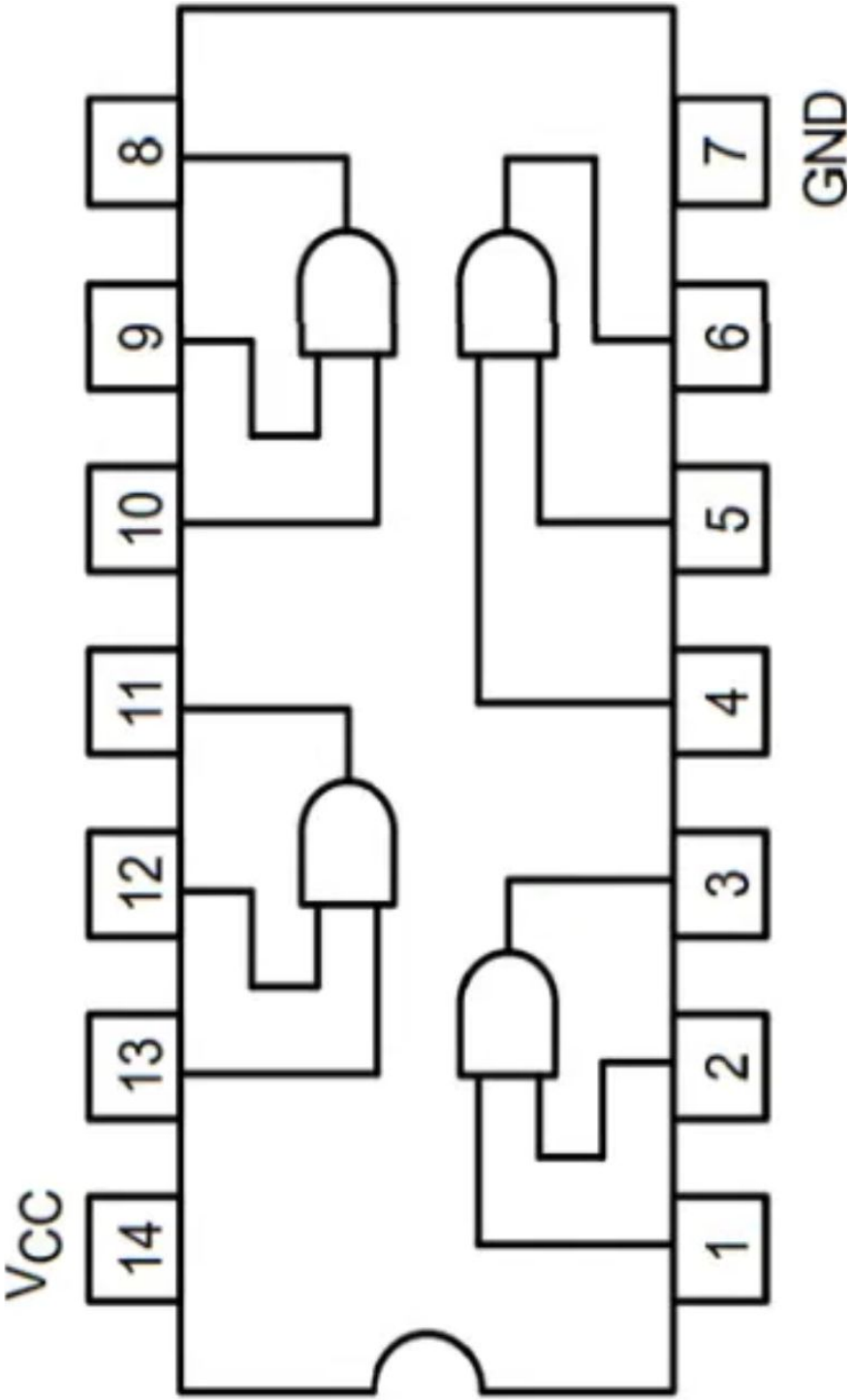
A presença de um pequeno cículo

Tabela-verdade

PORTA	SÍMBOLO GRÁFICO	TABELA-VERDADE	FUNÇÃO LÓGICA	EXPRESSÃO
E AND		A B S 0 0 0 0 1 0 1 0 0 1 1 1	Assume valor 1 quando todas as entradas forem 1 e 0 nos outros casos.	$S = A \bullet B$
OU OR		A B S 0 0 0 0 1 1 1 0 1 1 1 1	Assume valor 0 quando todas as entradas forem 0 e 1 nos outros casos.	$S = A + B$
NÃO NOT INVERSORA	 A presença de um pequeno cículo sempre indica ser uma inversora	A S 0 1 1 0	Inverte o valor da variável de entrada A	$S = \bar{A}$

74LS09 Quad 2-input AND Gate

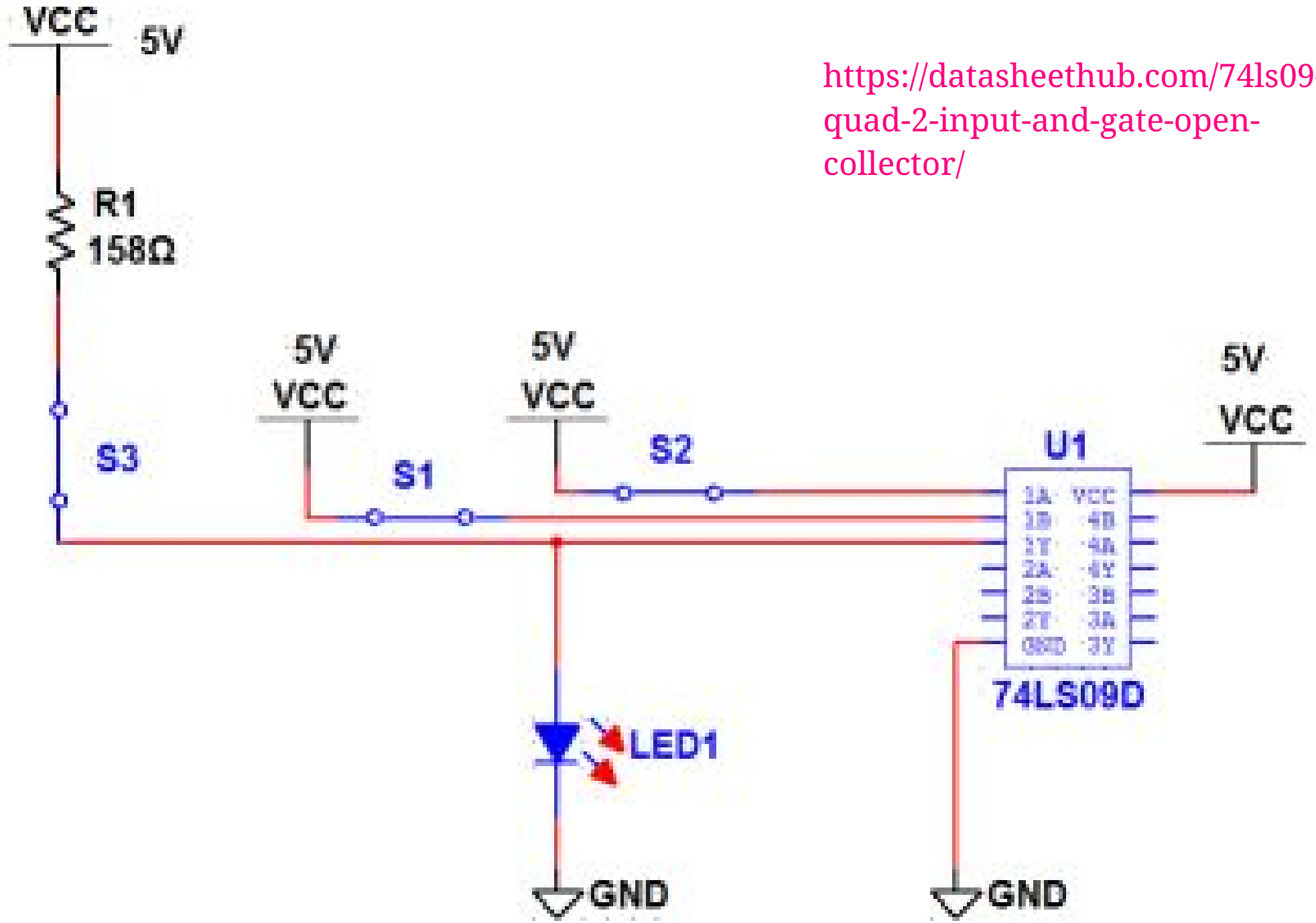




https://www.eeeshopbd.com/product/74ls09-2-input-and-gates

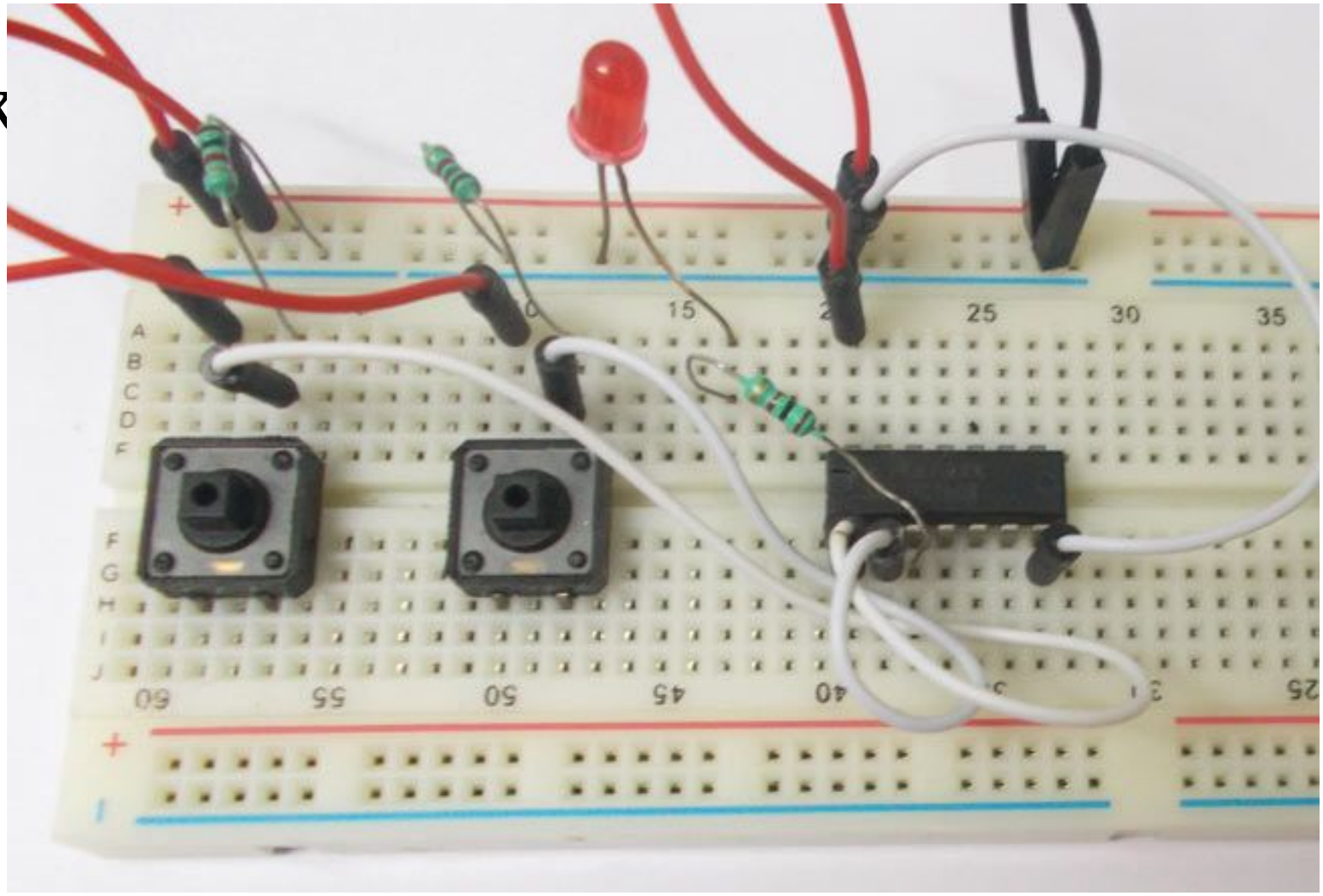
Circuito - exemplo

https://datasheethub.com/74ls09-quad-2-input-and-gate-open-collector/



Exemplo

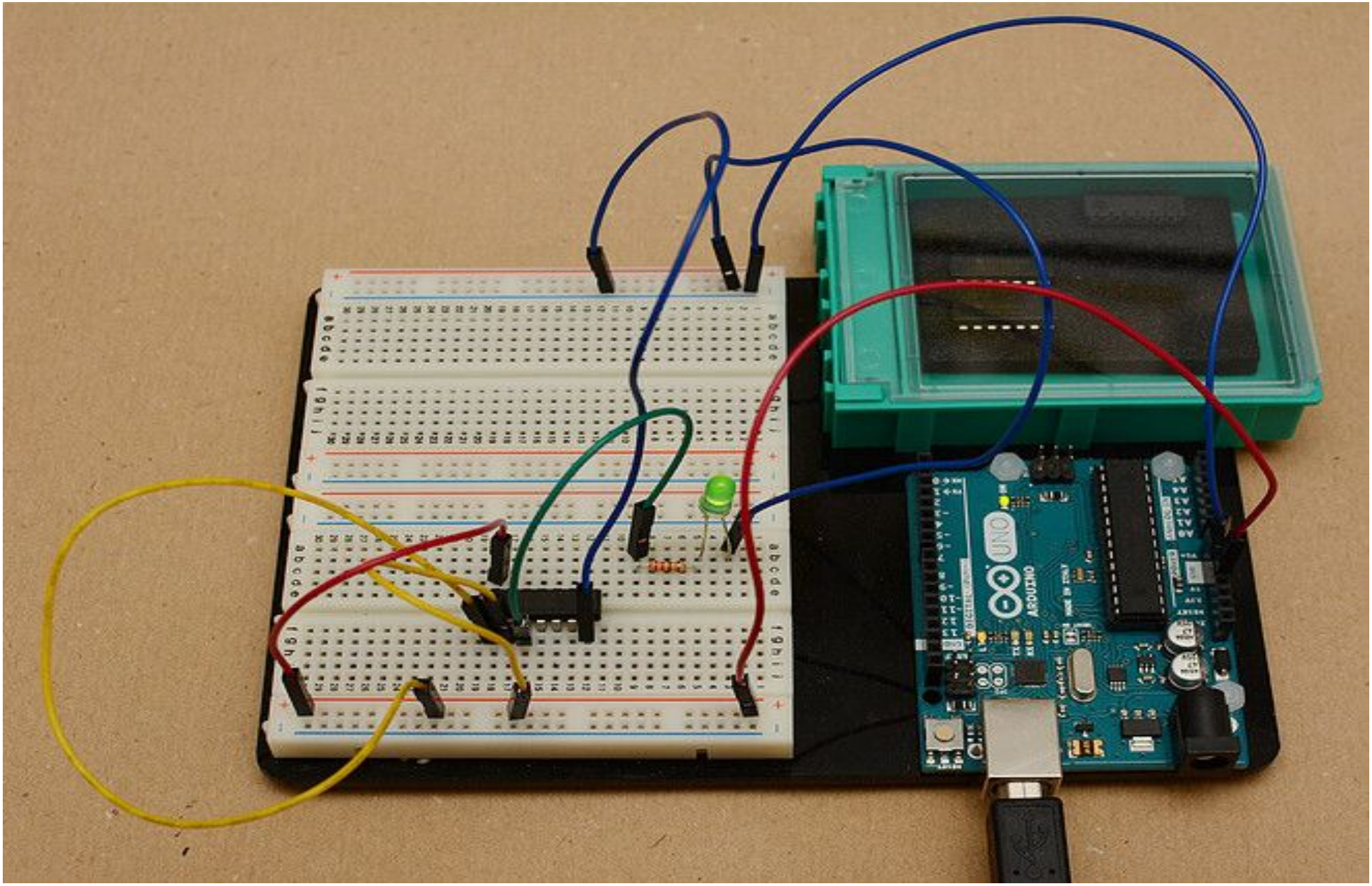
Click



https://youtu.be/rKt1LXkcrwE
https://www.build-electronic-circuits.com/7400-series-integrated-circuits/74hc09-74ls09

Exemplo

https://forum.arduino.cc/t/sn74ls08n-and-sn74ls05n-doesnt-work/427679/4



Famílias lógicas

TTL - *Transistor-Transistor Logic* (geralmente 5V)

Versões: 74, 74LS, 74S, 74AS, 74ALS

CMOS - *Complementay MOS* (várias faixas de tensão)

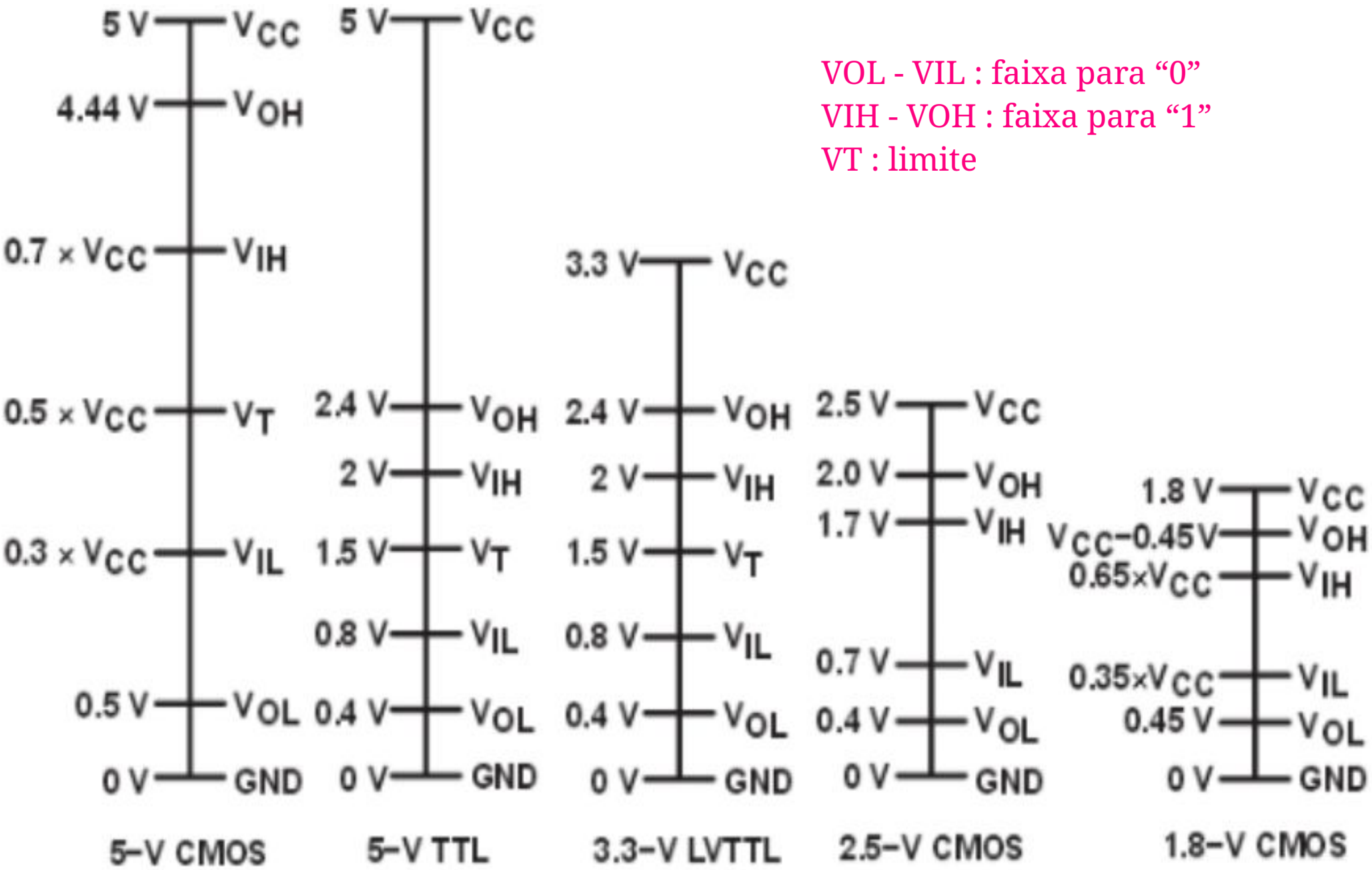
Versões: 4000, 74C, 74HC, 74HCT, 74LV, 74VC, 74AC, 74ACT

A “série 7400” surgiu em 1966, pela Texas Instruments

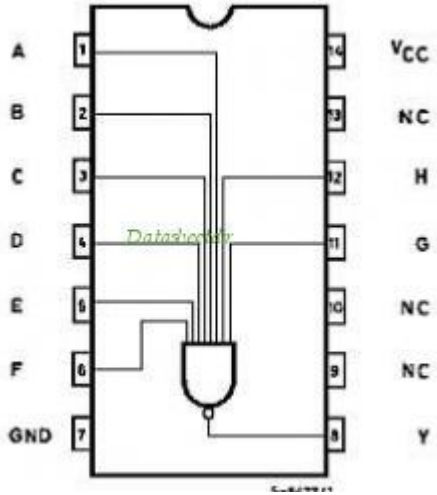
WENDLING, M. Prof. Famílias Lógicas I. https://www.feg.unesp.br/Home/PaginasPessoais/ProfMarceloWendling/6---familias-logicas-i---ii.pdf

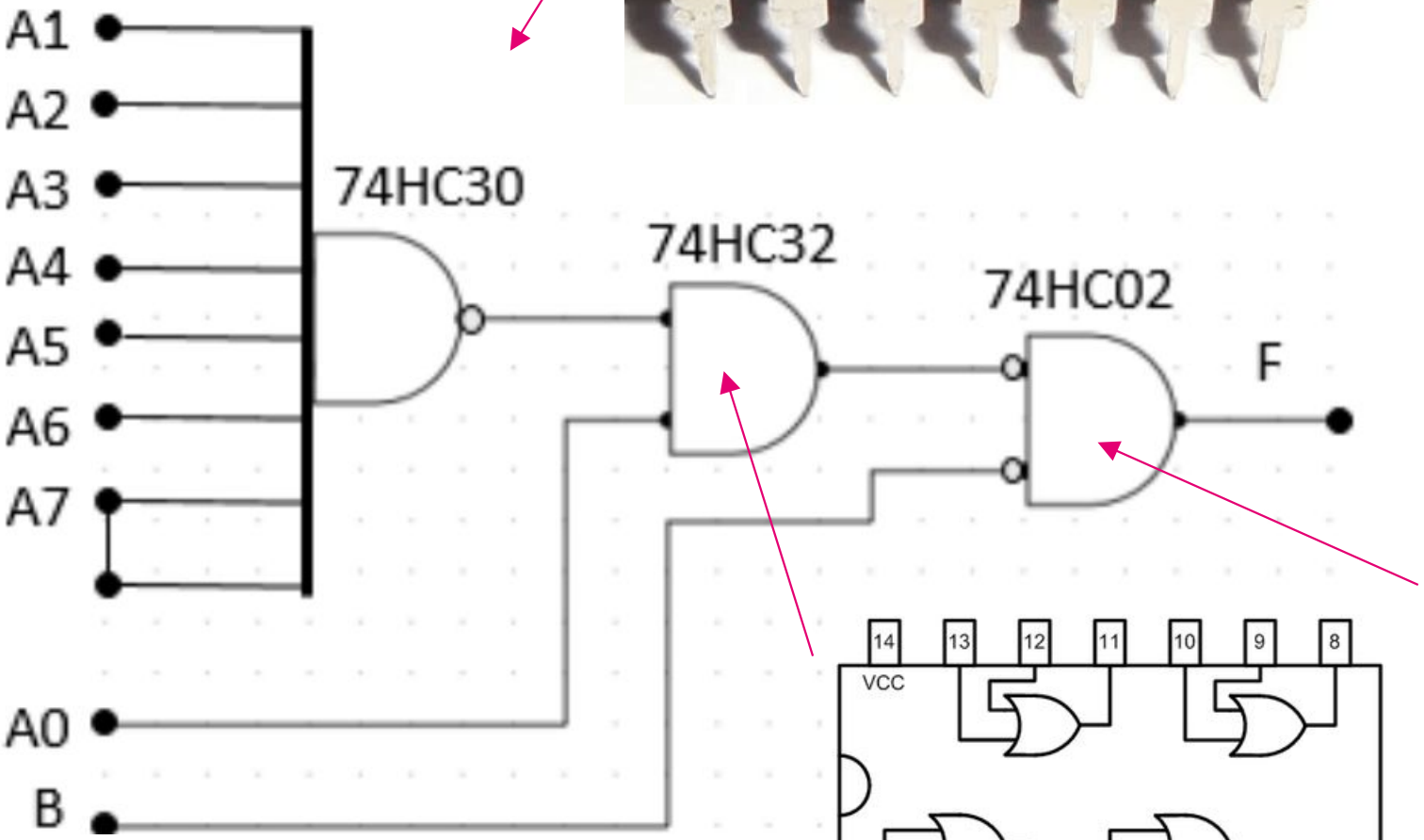
Níveis de tensão - alguns exemplos

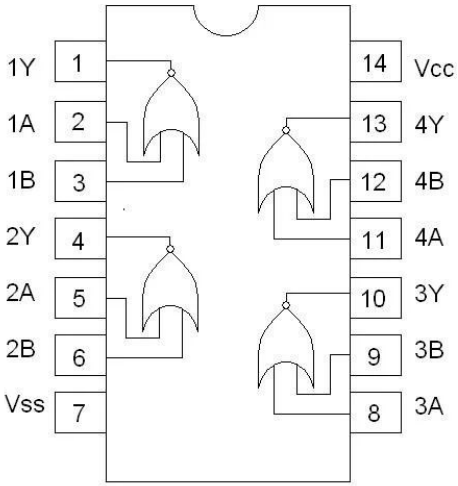
VOL - VIL : faixa para “0”
VIH - VOH : faixa para “1”
VT : limite

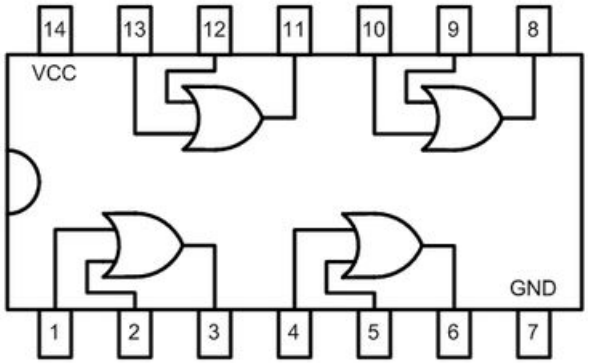


Arranjo de portas lógicas

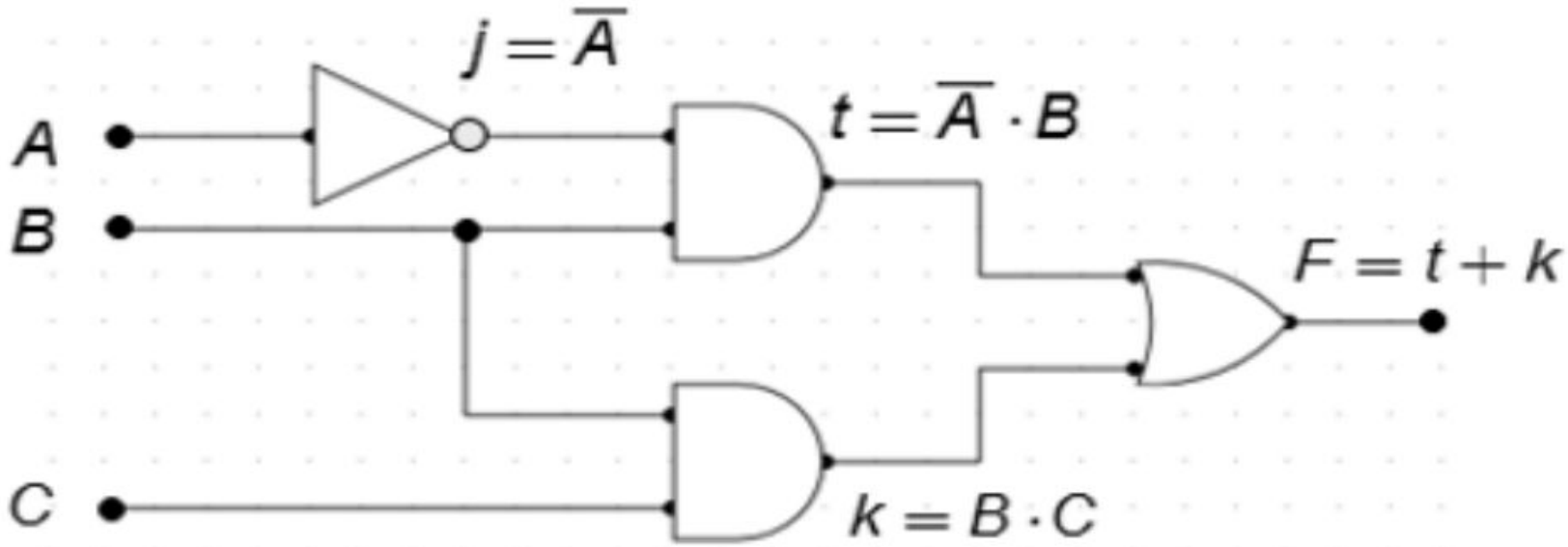






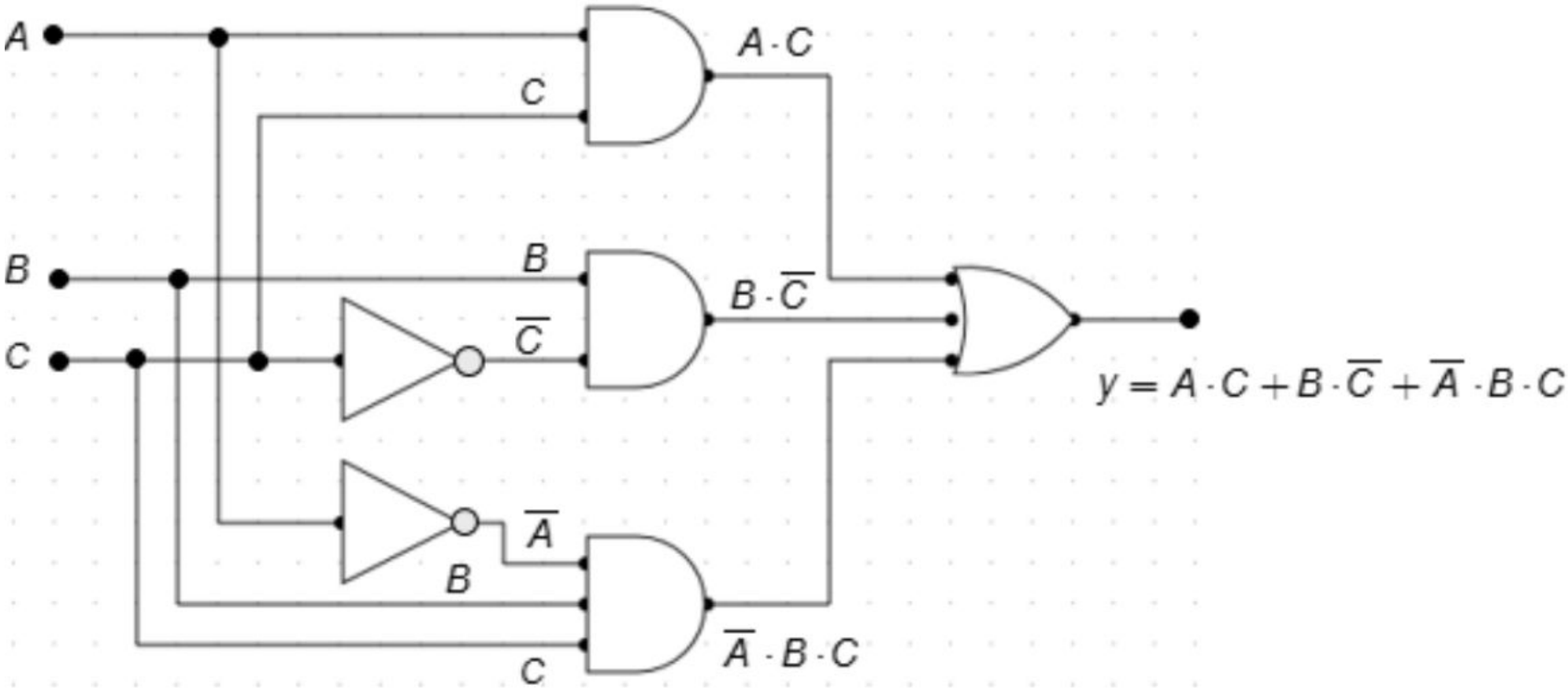


Circuito lógico completo



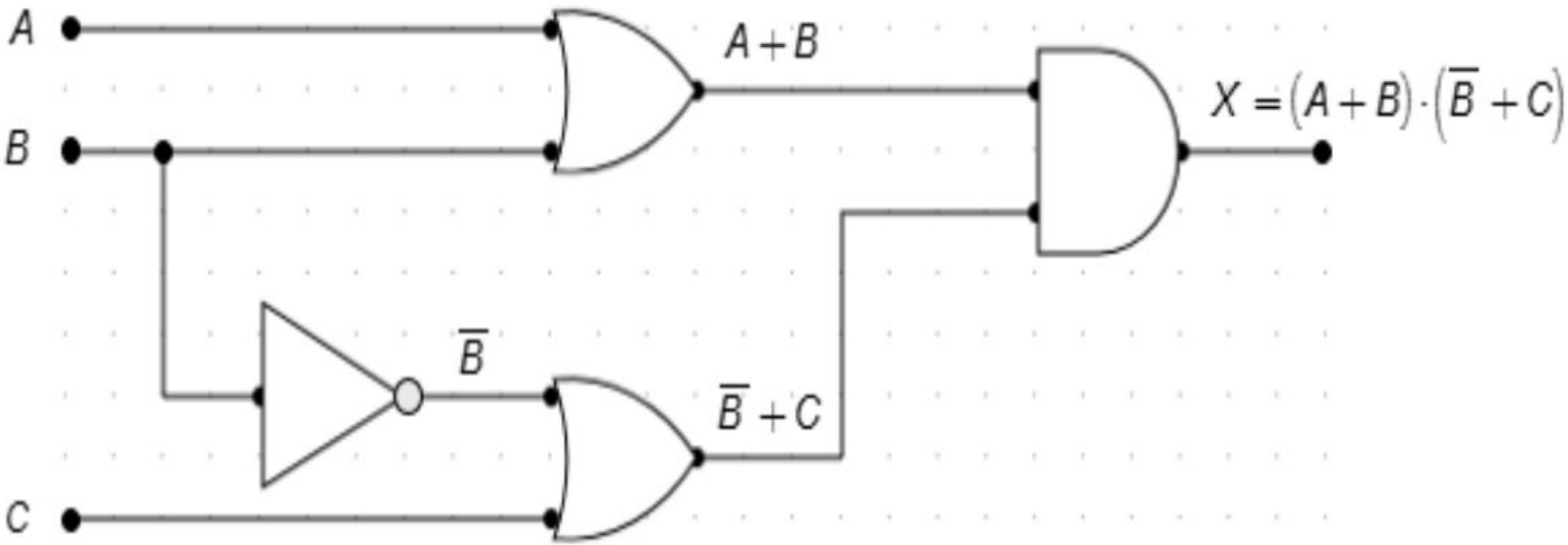
A	B	C	$J = \overline{A}$	$t = \overline{A}.B$	$k = B.C$	$F = t + k$
0	0	0	1	0	0	0
0	0	1	1	0	0	0
0	1	0	1	1	0	1
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	0	0	0
1	1	1	0	0	1	1

Arranjo de portas lógicas equivalentes

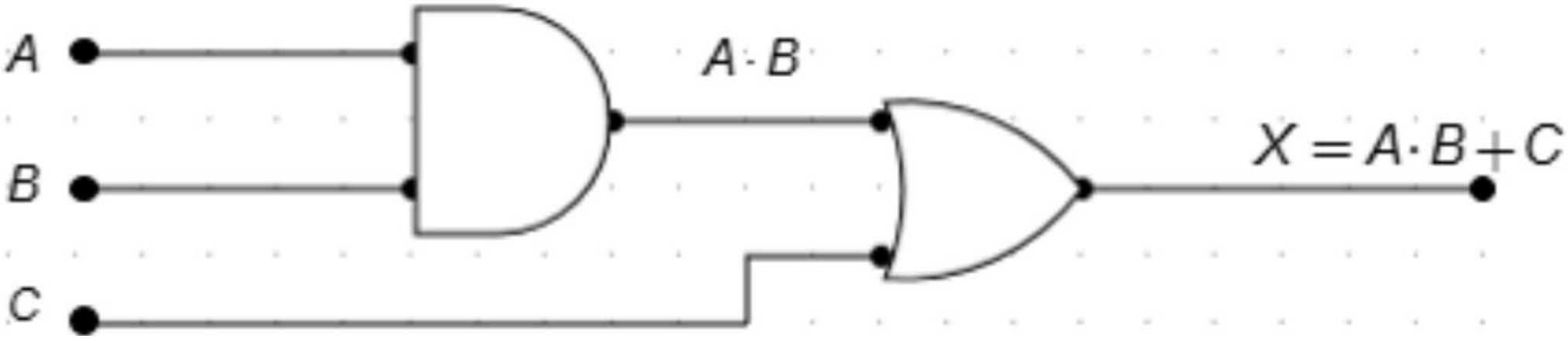


Obtenção do circuito

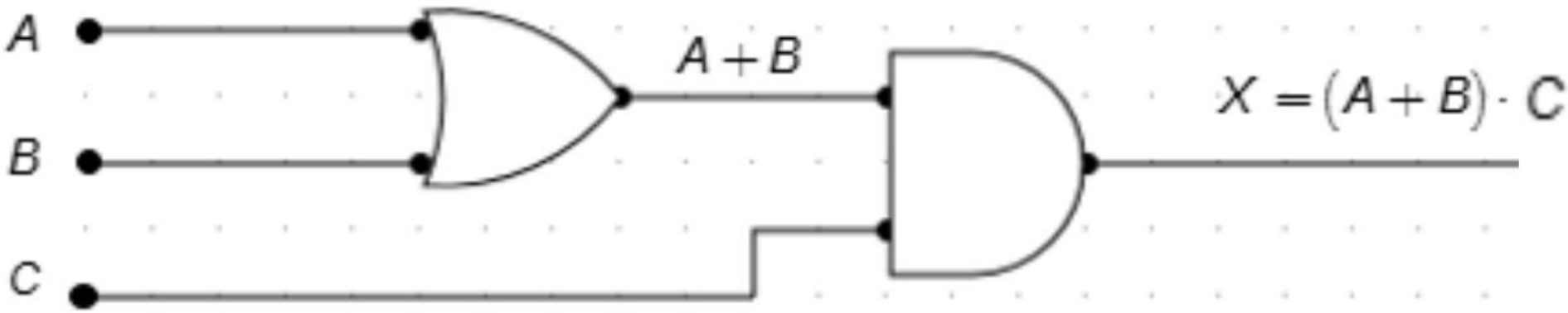
$$X = (A + B)(\overline{B} + C)$$



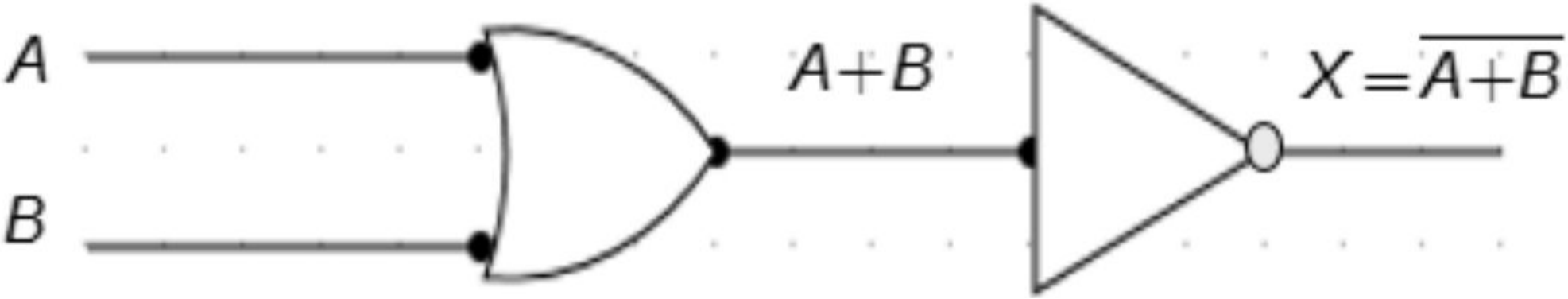
Obtenção da expressão booleana



Produto de uma soma

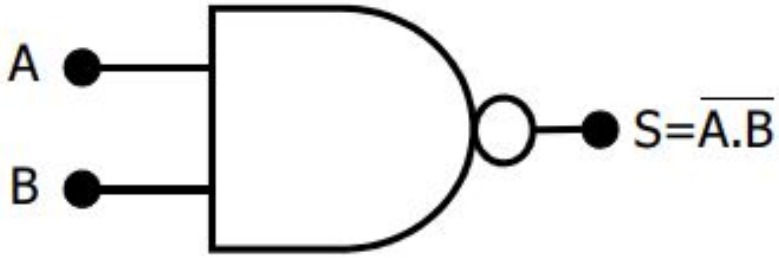


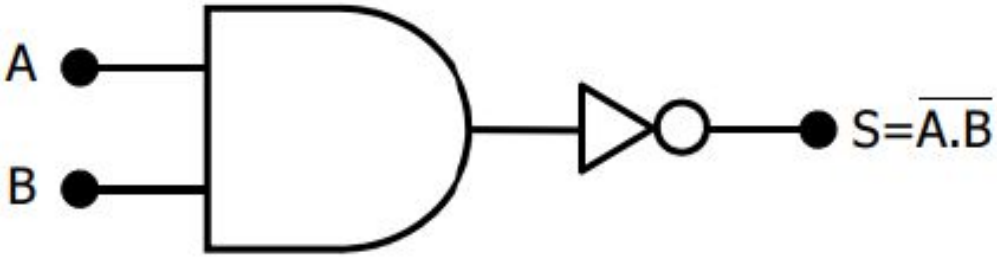
Inversor na saída



NOT com AND = NAND

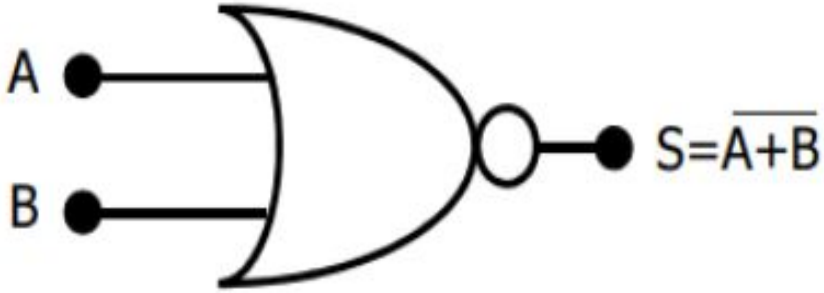
A	B	$S=\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

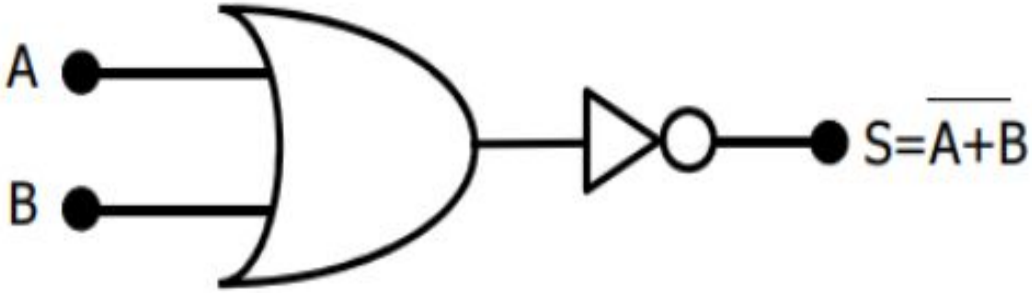




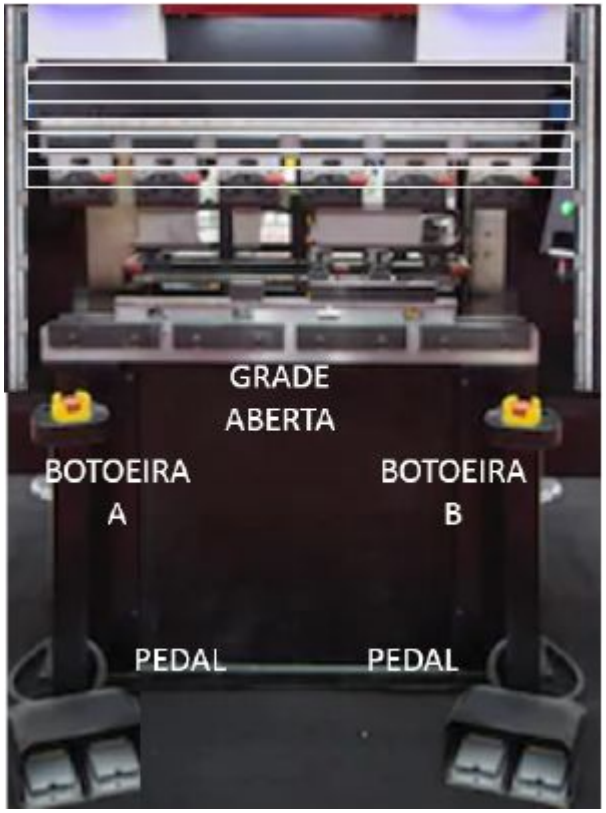
Função NOT com OR = NOR

A	B	$S=\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0





Exemplo - prensa



O sistema é ativado com nível lógico “1” e desativado com nível lógico “0”

O sistema ainda deve apresentar uma chave de “ligado ou desligado”, não considerada neste projeto

Ambas as botoeiras precisam ser acionadas simultaneamente

A grade de segurança deve ser a primeira variável a ser checada e ativada apenas na posição “grade fechada” e desativada na posição “grade aberta”

A prensa só será acionada se todos as entradas estiverem ativas

2 pedais de cada lado com acionamento (com o pé) único

Não há a necessidade de usar os dois pés para acioná-lo

Todas as entradas necessitam estar ativas (1) para a prensa funcionar

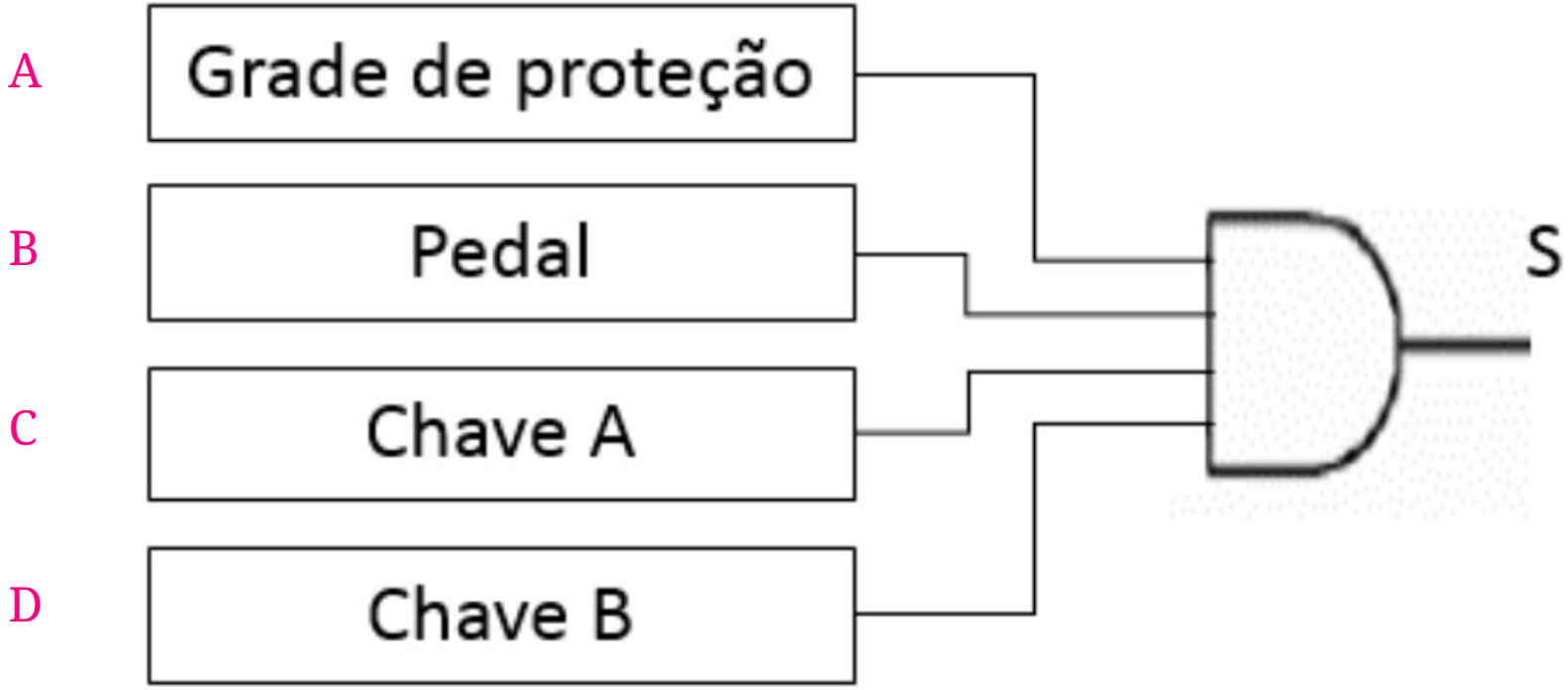
Tabela-verdade

A	B	C	D	
GRADE	PEDAL	BOTÃO A	BOTÃO B	SAÍDA
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

A equação booleana, portanto, seria:

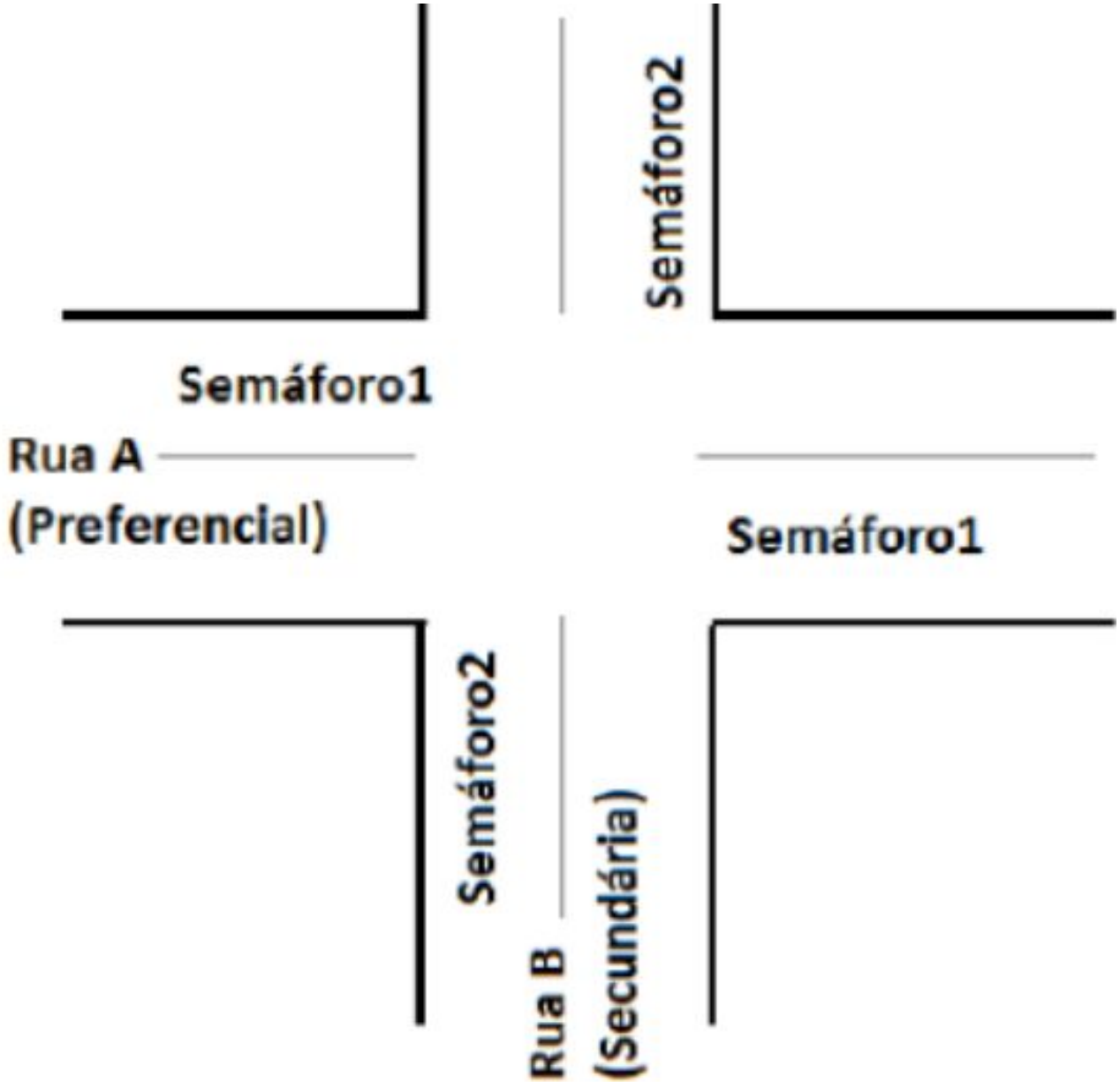
S = A . B. C .D

O circuito equivalente seria uma porta AND com 4 entradas:



Outro exemplo

Cruzamento de vias



Problema

Quando houver carros transitando somente na Rua B, o semáforo 2 deverá permanecer verde para que esses veículos possam trafegar livremente

Quando houver carros transitando somente na Rua A, o semáforo 1 dever permanecer verde pelo mesmo motivo

Quando houver carros transitando nas Ruas A e B, é preciso abrir o semáforo para a Rua A, pois esta é a preferencial

Haverá apenas as cores vermelho e verde nos semáforos

Entradas

Sensor de presença na rua A, registrando carro em A → A = 1; logo, não havendo carros em A, A = 0

Sensor de presença na rua B, registrando carro em B → B = 1; logo, não havendo carros em B, → B = 0

Saídas

Verde do semáforo 1 aceso → V1 = 1

Verde do semáforo 2 aceso → V2 = 1

Vermelho do semáforo 1 aceso → VM1 = 1

Vermelho do semáforo 2 aceso → VM2 = 1

Quando V1 = 1 temos: VM1 = 0, VM2 = 1, V2 = 0

Quando V2 = 1 temos: VM2 = 0, VM1 = 1, V1 = 0

Tabela-verdade

	A	B	V1	VM1	V2	VM2
Situação 1	0	0	0	1	1	0
Situação 2	0	1	0	1	1	0
Situação 3	1	0	1	0	0	1
Situação 4	1	1	1	0	0	1